Hu Zhentao

AI Application Engineer - Audio/Video Pipelines, LLM Agents, and Applied ML

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TECHNICAL PROFILE

- Applied AI builder with a combined background in AI product delivery, algorithm research, and engineering integration. Strongest depth is in audio/video task chains, digital-human voice interaction, AIGC production workflows, agent tool calling, and wireless sensing algorithms.
- Able to translate business scenarios into data, model, API, frontend, backend, deployment, and acceptance nodes: from microphone capture, VAD, ASR, LLM/Agent, and TTS to digital-human rendering, and from GPU queues and retries to subtitle/dubbing timeline alignment.
- Research foundation covers PyTorch, Transformer/CNN models, time-series signal processing, mmWave radar datasets, and benchmark reproduction. Engineering exposure includes FastAPI/React/Vite integration, static deployment, model/plugin integration, service rollout, debugging, and patching.

TECHNICAL SKILLS

- Audio and digital humans: Whisper, Llama3-8B, Tacotron2-DDC-GST, Qwen, DeepSeek, VAD, ASR/TTS, AEC/3A, OpenChatAvatar, Coze Agent, Doubao real-time voice.
- Video and AIGC pipelines: storyboard generation, image/video generation, GPU compute relay, task queues, status polling, failure retry, cost accounting, video translation, subtitle timeline alignment, dubbing synthesis, multilingual export.
- Algorithms and models: Python, PyTorch, Transformer, CNN, ResNet, Swin Transformer, ViT, SFT/PEFT, DPO/PPO/GRPO route understanding, mmWave radar, and time-series signal processing.
- Engineering delivery: FastAPI, React, Vite, API contracts, frontend/backend integration, task-state design, result-file handoff, static-site deployment, vendor/model-platform coordination, log analysis, debugging, and patches.

TECHNICAL WORK IN BUSINESS PROJECTS

Early-Stage Startup - Applied AI and Audio/Video Projects

Jan 2026 - Present

- Initiated and drove a wellness companion digital-human project, decomposing elderly companion scenarios into modules for conversation, memoir generation, local music, scenario Q&A, emergency fallback, family linkage, and privacy authorization. The project secured about RMB 800K in staged budget support and moved into commercial-demo and pilot discussions.
- Built a self-hosted digital-human interaction demo with OpenChatAvatar, connecting microphone capture, VAD, ASR, LLM/Agent reasoning, TTS, audio playback, and rendering-state synchronization so ASR, reasoning, synthesis, and business-agent modules can be replaced later.
- Designed a Coze-agent route where the local audio chain handles capture, recognition, synthesis, and playback, while Coze Agent owns companion dialogue, intent recognition, tool routing, knowledge base, and business logic to reduce configuration cost.
- Integrated a Doubao end-to-end real-time voice route, wrapping streaming input/output, interruption, silence detection, turn state, and digital-human speaking/rendering interfaces to validate lower-latency dialogue compared with a serial ASR-LLM-TTS chain.
- Advanced AI comic and video-translation workflows, structuring the path from content/IP acquisition to storyboard generation, image/video generation, GPU task scheduling, multilingual translation, and distribution monetization. Partner-side pilot and procurement interest has been expressed.
- Designed a compute-relay scheme for generation tasks, covering GPU node access, task submission, queue scheduling, status polling, result-file return, permission control, failure retry, and cost accounting to decouple frontend requests from model runtime environments.
- Designed a video-translator chain connecting audio extraction, ASR transcription, translation polishing, subtitle timeline alignment, dubbing synthesis, and multilingual export, with explicit inputs for source video, subtitles, character voice, target language, segmentation, output format, and manual review points.
- Developed AIGC video workflows covering script planning, anchor-image generation, segmented video generation, programmatic evaluation, failure retry, and resumable execution, supporting e-commerce virtual try-on and semi-automated digital-human templates.
- Contributed to a multi-agent business advisory assistant using function calling and prompt engineering to coordinate product introduction, Q&A, recommendation, lead capture, and tool-calling flows for client demos and scenario adaptation.
- Worked on model-platform, agent-tool, and digital-human hardware coordination; prepared project-solution, staged-budget, procurement, vendor, and deployment materials; and handled integration and issue patches around service interfaces, runtime state, media assets, and delivery acceptance.

Huawei Consumer BG - Audio Algorithm Engineer Intern

Jul 2024 - Oct 2024

- Studied spectrograms, 3A algorithms, AEC evaluation, and endpoint audio processing. Based on ICASSP 2021 nn-ref related work, built a research prototype and collected experimental data.
- Participated in endpoint AEC testing on iPhone and Huawei devices, gaining hands-on experience in audio algorithm evaluation, device-side comparison, issue localization, and test documentation.

ALGORITHM FOUNDATION AND APPLIED RESEARCH

Speech Interaction Model Based on Llama3-8B

Oct 2024 - Dec 2024

- Built an ASR-LLM-TTS pre-research chain with whisper_large_v2, Llama3-8B, and tacotron2-DDC-GST to validate speech recognition, dialogue reasoning, and speech synthesis integration.
- Tested Qwen and DeepSeek model families on audio-data fine-tuning scenarios, comparing parameter-efficient fine-tuning with full-parameter fine-tuning to inform later audio-agent and digital-human voice-route choices.

Human Activity Recognition with mmWave Signals

Jun 2023 - Dec 2025

- Collected raw mmWave radar signals, organized datasets, defined the experimental plan, and completed thesis experiments. The dataset covered 6 daily activity classes and 20 participants, exceeding the public datasets used as references at the time.
- Reproduced and compared ResNet, Swin Transformer, ViT, and related variants to build a multi-dimensional benchmark with fully reproduced baselines and cross-model comparisons.
- Combined traditional signal processing inspired by wavelet transform and fractional Fourier transform with Transformer/CNN components to improve generalization and task adaptability for end-to-end activity recognition.
- Explored sensor-data question answering and reasoning by organizing preprocessed mmWave outputs into QA data, using Qwen2.5-32B inference and Qwen2.5-1.5B SFT, and comparing DPO/PPO/GRPO-style routes for recognition and generation behavior.

EDUCATION

University of Chinese Academy of Sciences - M.S. in Electronic Information, Institute of Computing Technology

Sep 2022 - Jan 2026

- Research interests: artificial intelligence, sensor algorithms, time-series signal processing, and LLM applications.

Xidian University - B.S. in Electronic Information

Sep 2018 - Jun 2022

- Foundation: image processing, signal processing, signals and systems, random signal analysis, electromagnetic fields, and wireless-system courses.

PUBLICATIONS

- Time-domain multi-scale analysis based wireless human activity recognition method, first author; inspired by wavelet transform and using a Transformer-based preprocessing module.
- T-FrFT: Fractional Fourier Transform based Discriminative Feature Extraction Method for Human Activity Classification, IJCNN 2025, first author.